

Total no. of Pages: 1

Roll no.

Degree: B.Tech (CSDA / CSIOT)

Semester: 2nd semester**MID-SEMESTER EXAMINATION, Feb. 2026**

Course Title: DIGITAL LOGIC DESIGN

Course Code: CBEPC203 / CIEPC203 (NEP SCHEME)

Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Given decimal numbers $X = 49.125$ and $Y = 34.625$ Find their Binary and Hexadecimal equivalent values.	2	1
1b	Evaluate binary sum and difference of X and Y in the above question (1a).	2	1
2a	If $A = 632$ and $B = 368$, find the BCD sum of A and B.	2	1
2b	Implement logic circuit, using <i>minimum NAND gates</i> for: $F(W, X, Y, Z) = \sum_{min} (0, 6, 10, 14) + d(2, 8, 11, 12)$	2	1
3a	Design a comparator logic circuit for $A_1 A_0$ and $B_1 B_0$ inputs.	2	1, 2
3b	Draw the logic circuit of a 4-bit BCD adder.	2	2
4a	$F(P, Q, R, S) = \sum_{min} (0, 1, 3, 4, 7, 12, 13, 15)$ Implement using a single 4:1 Multiplexer.	2	2
4b	Design a 8:1 MUX, using 4:1 MUX.	2	2
5a	Design a logic circuit to detect invalid Excess-3 inputs.	2	2
5b	Given $F_1(W, X, Y) = \sum_{min} (1, 3, 4, 5, 7)$ and $F_2(W, X, Y) = \sum_{min} (0, 1, 2, 4, 7)$ Implement both F_1 & F_2 using a suitable De-MUX.	2	2

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Roll no.

Degree: B.Tech (CSDA / CSIOT)

Semester: 2nd semester**MID-SEMESTER EXAMINATION, Feb. 2026**

Course Title: DIGITAL LOGIC DESIGN

Course Code: CBEPC03 / CIEPC03 (OLD SCHEME)

Duration: 1:30 Hours

Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Given decimal numbers $X = 49.125$ and $Y = 34.625$ Find their Binary and Hexadecimal equivalent values.	1.5	1
1b	Evaluate binary sum and difference of X and Y in the above question (1a).	1.5	1
2a	If $A = 632$ and $B = 368$, find the BCD sum of A and B.	1.5	1
2b	Implement logic circuit, using <i>minimum NAND gates</i> for: $F(W, X, Y, Z) = \sum \min (0, 6, 10, 14) + d(2, 8, 112)$	1.5	1
3a	Design a comparator logic circuit for $A_1 A_0$ and $B_1 B_0$ inputs.	1.5	1, 2
3b	Draw the logic circuit of a 4-bit BCD adder.	1.5	2
4a	$F(P, Q, R, S) = \sum \min (0, 1, 3, 4, 7, 12, 13, 15)$ Implement using a single 4:1 Multiplexer.	1.5	2
4b	Design a 8:1 MUX, using 4:1MUX.	1.5	2
5a	Design a logic circuit to detect invalid Excess-3 inputs.	1.5	2
5b	Given $F_1(W, X, Y) = \sum \min (1, 3, 4, 5, 7)$ and $F_2(W, X, Y) = \sum \min (0, 1, 2, 4, 7)$ Implement both F_1 & F_2 using a suitable De-MUX.	1.5	2